

Aadhitya Balaji

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EDUCATION

University of Wisconsin-Madison | Madison, WI

Expected Graduation Date – May 2029

Bachelor of Science: Double Major in Computer Science and Data Science

- Minor in Business
- GPA: 3.92/4.0
- Dean's List: Fall '25 & Spring '26

TECHNICAL SKILLS

LANGUAGES: Python, Java, JavaScript, R, SQL, HTML/CSS

MACHINE LEARNING & DATA SCIENCE: PyTorch, TensorFlow, OpenCV, Pandas, NumPy

CLOUD & DEVOPS: AWS (S3, Lambda, EC2, ECR, VPC), Docker, Git

DATABASES, WEB DEVELOPMENT & TOOLS: Postgres, VS Code, React, Firebase

AI SKILLS: Claude Code, GitHub Copilot, Cursor, MCP

TECHNICAL PROJECTS AND RESEARCH

AI Pickleball Coach | Early Venture

April 2026 - Present

Lead Software & Cloud Engineer

- **Partnered with a commercial pickleball facility** and built a personalized coaching application that extracts biomechanical metrics and generates player-specific recommendations via the Claude API
- **Cut video processing time 3x** by containerizing the pipeline with Docker and deploying it via AWS Lambda, ECR, and EC2 instances for automated ingestion and inference
- **Improved rally detection accuracy 20%** by fitting gravity-parabola trajectories to 2D ball tracks, recovering true court position via 3D physics-based reconstruction
- **Boosted ball-detection coverage 68x** (0.5% → 34% of frames) by fine-tuning TrackNetV2 on a custom human-labeled ball-position dataset

UW-Madison Computational Biology and Machine Learning Lab | Madison, WI

September 2025 - Present

Undergraduate Researcher

- **Designed MORFEA**, an unsupervised CNN-LSTM autoencoder modeling embryo development as latent trajectories, training on 700+ time-lapse sequences for label-free embryo quality classification
- **Built an initial feed-forward classifier** on MORFEA's latent embryo embeddings as a baseline for predicting developmental stage (cleavage, morula, blastocyst), informing later architecture decisions
- **Co-authored a manuscript on MORFEA**, invited for submission into a Frontiers Research Topic on deep learning-driven medical imaging

PROFESSIONAL AND LEADERSHIP EXPERIENCE

Wisconsin Consulting Club | Madison, WI

February 2026 – Present

Analyst | YouTube Music x Wisconsin Consulting Club Team

- **Conducted a deep-dive analysis of 5 competitor platforms**, pulling usage metrics and visualizing conversion funnels to surface strategic gaps to improve user retention and increase the customer base
- **Designed mockups for 2 feature deep-dives** and created a user journey map, translating research into 8 feature recommendations that shaped 2 of the team's 4 final strategic deliverables

Leadership Initiatives | Washington D.C.

July 2023 – May 2025

Public Health Staff Intern

- **Directed 9 cross-functional intern teams** consisting of up to 63 total researchers to deploy 11 targeted public health campaigns across workshops in Bauchi State, Nigeria